

DATA SCIENCE TOOL BOX: PYTHON PROGRAMING
PROJECT REPORT

(Project Semester January-June 2026)

Indian Census Data Analysis using Python

Submitted by

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Programme and Section – B. Tech and 224YM

Course Code – INT375

Under the Guidance of

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Discipline of CSE/IT

Lovely School of computer Science and Engineering

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CERTIFICATE

This is to certify that Gaurav Raj bearing 12400788 Registration no. has completed INT375 project titled, Indian Census Data Analysis using Python Dashboard under my guidance and supervision. To the best of my knowledge, the present work is the result of my original development, effort and study.

Signature and Name of the Supervisor

Designation of the Supervisor

School of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab.

Date: 11-04-2026

DECLARATION

I, Gaurav Raj student of Bachelor of Technology under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 11-04-2026

Signature – Gaurav Raj

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Name of the student:
Gaurav Raj

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Dr. Baljinder Kaur** mam for their valuable guidance, support, and encouragement throughout the completion of this Data Science Minor Project.

I am also thankful to Lovely Professional University for providing the necessary resources and academic environment to successfully carry out this work.

Gaurav Raj

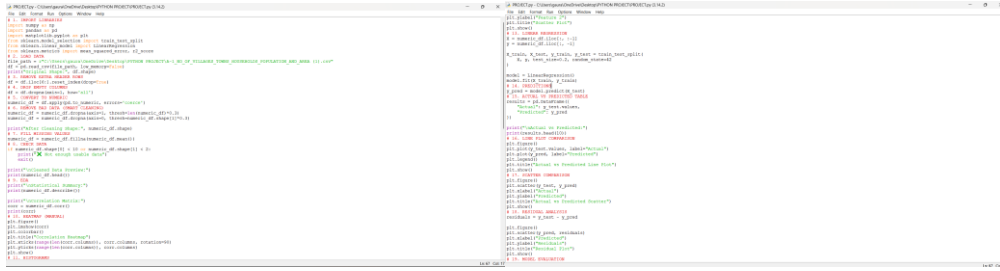
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INTRODUCTION

A screenshot of a Jupyter Notebook interface with two cells of Python code. The left cell contains code for data loading and preprocessing, including imports for pandas, numpy, and matplotlib, and a function to load census data. The right cell contains code for data analysis, including a function to calculate the population distribution by state and a function to calculate the literacy rate by state.

(Indian Census Data Analysis using Python)

In the modern era, data plays a critical role in decision-making processes across industries. Governments rely heavily on demographic data to design policies, allocate resources, and plan infrastructure.

This project uses **Python-based data science techniques** to analyze census-related data. The focus is on:

- Understanding population distribution
- Identifying relationships between demographic variables
- Applying machine learning for prediction

The project simulates a real-world analytical workflow including data preprocessing, analysis, and modeling.

OBJECTIVES

Analytical Objectives

- To clean and preprocess raw data
- To perform exploratory data analysis
- To understand statistical properties of data

Technical Objectives

- To visualize data using Python libraries
- To build a regression model
- To evaluate model performance

Practical Objectives

- To derive insights useful for planning and decision-making
- To understand real-world data challenges

The objectives of this project focus on analyzing and understanding the dataset through both analytical and technical approaches. The first objective is to clean and preprocess the raw data to make it suitable for analysis by removing inconsistencies and handling missing values. The project also aims to perform exploratory data analysis (EDA) to identify patterns, relationships, and overall data behavior, along with understanding basic statistical properties like mean and variation. From a technical perspective, the objective is to visualize the data using Python libraries and build a Linear Regression model to study the relationship between variables. The performance of the model is evaluated using standard metrics to assess its accuracy. Finally, the project aims to derive meaningful insights that can support real-world decision-making and to gain practical experience in handling real-world data challenges.

DATASET DESCRIPTION

DATASET DESCRIPTION

The dataset contains demographic information including:

- Number of villages
- Number of towns
- Number of households
- Population
- Area

After cleaning:

- **Original Shape:** (20022, 86)
- **Cleaned Shape:** (19989, 8)

The dataset is numerical and suitable for statistical and machine learning analysis.

The dataset contains demographic information such as the number of villages, towns, households, population, and area. Initially, it consisted of 20022 rows and 86 columns, but after preprocessing and cleaning, it was reduced to 19989 rows and 8 relevant columns. The final dataset is primarily numerical, making it suitable for statistical analysis and machine learning applications.

DATA PREPROCESSING

Steps performed

- Removed extra header rows
- Dropped empty columns
- Converted data into numeric format
- Removed columns with excessive missing values
- Removed rows with insufficient data

- Filled missing values using mean

This ensured the dataset became clean, structured, and usable

Data preprocessing is one of the most critical steps in any data science project.

✓ Step 1: Removing Extra Header Rows

The dataset contained unnecessary header rows which were removed using:

```
df = df.iloc[6:]
```

✓ Step 2: Dropping Empty Columns

Columns with all missing values were removed:

```
df.dropna(axis=1, how='all')
```

✓ Step 3: Converting to Numeric

All values were converted to numeric:

```
pd.to_numeric(errors='coerce')
```

✓ Step 4: Handling Missing Values

- Columns with less than 30% data were removed
- Rows with insufficient values were dropped

✓ Step 5: Filling Missing Data

Remaining missing values were replaced using mean:

```
fillna(mean)
```

TOOLS & TECHNOLOGIES

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

The project was carried out using Python as the primary programming language due to its flexibility and wide range of data science libraries. Pandas was used for data cleaning, manipulation, and handling structured datasets, while NumPy supported efficient numerical computations. Matplotlib was utilized to create visualizations such as histograms, scatter plots, and heatmaps, which helped in understanding data patterns and relationships. Scikit-learn was used to implement the Linear Regression model and evaluate its performance

using metrics like Mean Squared Error (MSE) and R^2 score. Together, these tools enabled efficient data analysis, visualization, and predictive modeling.

EXPLORATORY DATA ANALYSIS (EDA)

EDA helps understand the structure and behavior of data.

Statistical Summary

Key observations:

- Mean values indicate average distribution
- Standard deviation shows variation
- Large difference between min and max → presence of outliers

🔍 Insight:

The dataset is **highly varied and not uniformly distributed**

Distribution Analysis (Histograms)

🔍 **Observations:**

- Most variables are **right-skewed**
- High frequency of low values
- Few extreme high values

🔍 Meaning:

- Data is not normally distributed
- Outliers exist

Exploratory Data Analysis (EDA) was performed to understand the behavior and characteristics of the dataset. Statistical measures such as mean, standard deviation, and range were analyzed, revealing significant variation and the presence of outliers. Distribution analysis using histograms showed that most variables are right-skewed, with values concentrated in lower ranges. This indicates that the data is not normally distributed and contains extreme values.

CORRELATION ANALYSIS

Correlation measures the relationship between variables.

Key Findings:

- Some features have **strong positive correlation**
- Some have **weak or negative correlation**

Example Insight:

- Feature 1 & Feature 2 → Strong relationship
- Feature 6 → Moderate influence

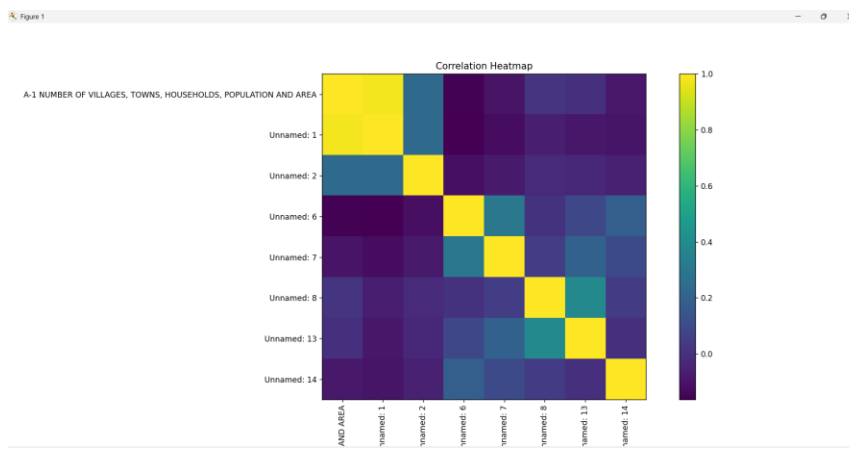
Importance:

Helps in:

- Feature selection
- Understanding dependencies

Correlation analysis was performed to examine the relationships between variables in the dataset. The results showed that some features have a strong positive correlation, while others exhibit weak or negative relationships. This helps in identifying important variables, understanding dependencies among features, and selecting relevant inputs for building an effective predictive model.

DATA VISUALIZATION



Heatmap

Shows correlation visually.

 Bright colour = strong relation

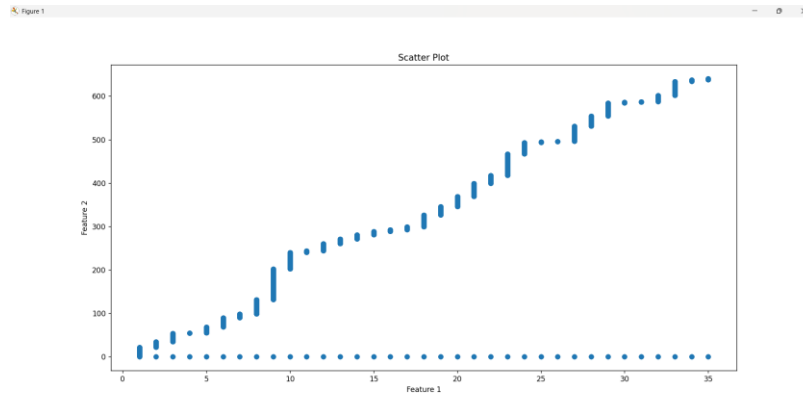
 Dark colour = weak relation

Histogram

Shows data distribution.

👉 Skewed distribution → presence of outliers

Scatter Plot



Shows relationship between variables.

👉 Upward trend = positive correlation

Data visualization techniques were used to better understand the dataset. The heatmap provided a visual representation of correlations, where brighter colors indicated stronger relationships and darker colors indicated weaker ones. Histograms revealed skewed distributions, suggesting the presence of outliers, while scatter plots showed relationships between variables, with upward trends indicating positive correlation.

MODEL EVALUATION

Mean Squared Error (MSE)

- Measures prediction error
- Lower value → better model

R² Score

- Shows how much variance is explained

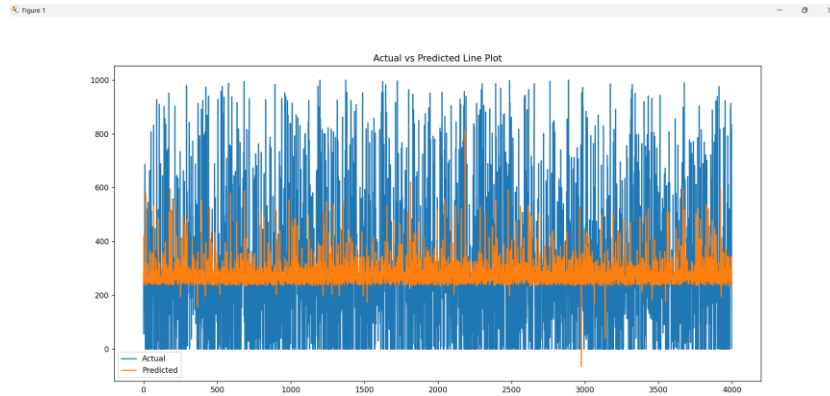
Interpretation:

- 0 → poor model
- 1 → perfect model

👉 Your Model:

- Moderate performance

RESULT ANALYSIS



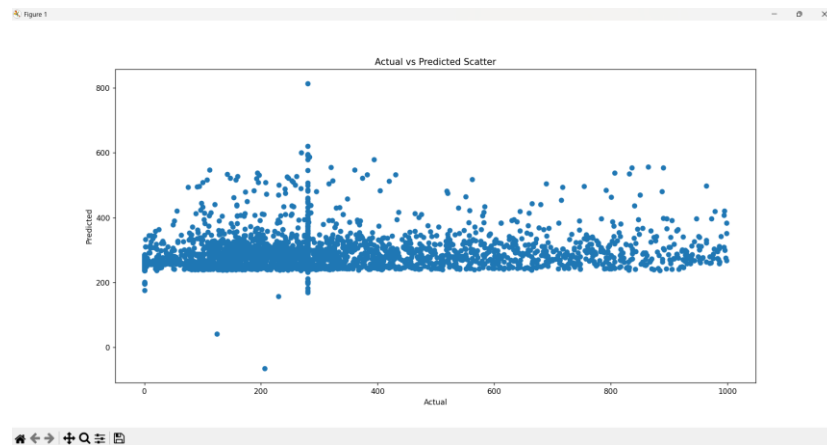
☒ Actual vs Predicted (Line Plot)

Observation:

- Actual values fluctuate heavily
- Predicted values are smoother

👉 Insight:

Model cannot capture extreme variations



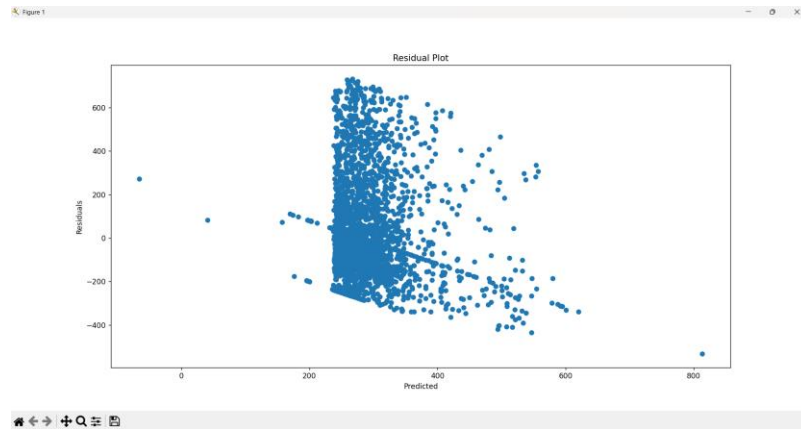
☒ Scatter Plot (Actual vs Predicted)

Observation:

- Points are widely scattered
- Not aligned on straight line

👉 Insight:

Weak predictive accuracy



Residual Plot

Observation:

- **Residuals are not randomly distributed**
- **Pattern exists**

👉 Insight:

Model assumption violated → non-linearity

BUSINESS / REAL-WORLD INSIGHTS

- **Population distribution is uneven**
- **Some regions dominate in metrics**
- **Strong relationships exist between variables**

👉 Applications:

- **Government planning**
- **Urban development**
- **Infrastructure allocation**

The analysis shows that population distribution is uneven, with certain regions contributing more significantly to key metrics. Strong relationships between variables were also observed, which can support better decision-making. These insights are useful for applications such as government planning, urban development, and infrastructure allocation.

CHALLENGES FACED

- Handling large dataset
- Missing values
- Irrelevant columns
- Model limitations

The project faced several challenges during its implementation. One of the primary difficulties was handling a large dataset, which required careful processing and increased computational effort. Managing missing values was another major challenge, as incomplete data can affect the accuracy and reliability of analysis; therefore, appropriate techniques like removal and mean imputation were applied. The presence of irrelevant and empty columns further complicated the preprocessing stage, requiring careful filtering to retain only useful features. Additionally, the limitations of the Linear Regression model posed a challenge, as it assumes a linear relationship between variables, whereas the dataset exhibited non-linear patterns and outliers, leading to moderate prediction accuracy.

LIMITATIONS

- Linear model assumption
- No categorical features
- No advanced models

The project has certain limitations that affect the overall performance and scope of the analysis. One major limitation is the use of a Linear Regression model, which assumes a linear relationship between variables, whereas the dataset exhibits non-linear patterns and variability. Additionally, the analysis is based only on numerical features, as no categorical variables were included, which may limit the depth of insights. Another limitation is the absence of advanced machine learning models such as Random Forest or XGBoost, which are better suited for handling complex data patterns and could potentially improve prediction accuracy. These limitations suggest that while the project provides valuable insights, there is scope for further improvement using more advanced techniques.

CONCLUSION

The project successfully demonstrates:

- ✓ Data preprocessing
- ✓ Visualization
- ✓ Regression modeling

However:

✍ Linear Regression is not sufficient

The project successfully demonstrates the complete data science workflow, including data preprocessing, exploratory data analysis, visualization, and regression modeling using Python. Through systematic cleaning and transformation, the raw dataset was converted into a structured and usable form. Various visualization techniques helped in understanding patterns, relationships, and distributions within the data. The Linear Regression model was applied to predict outcomes and evaluate relationships between variables. However, the results indicate that the model has only moderate accuracy due to the presence of non-linear patterns, outliers, and complex relationships in the dataset. This highlights that while Linear Regression serves as a good baseline model, it is not sufficient for capturing the full complexity of the data, suggesting the need for more advanced modeling techniques.

FUTURE SCOPE

- Random Forest
- XGBoost
- Neural Networks
- Feature engineering

The future scope of this project involves enhancing the model's accuracy and analytical depth by using more advanced techniques. Machine learning algorithms such as Random Forest and XGBoost can be applied, as they are capable of capturing complex and non-linear relationships in the data more effectively than Linear Regression. Additionally, Neural Networks can be explored to model deeper patterns and interactions between variables, especially in large and complex datasets. Another important improvement area is feature engineering, which involves creating new meaningful features or transforming existing ones to better represent the data. This can significantly improve model performance and provide deeper insights. By incorporating these advanced methods, the project can be extended to achieve higher predictive accuracy and more robust real-world applications.

REFERENCES

- Python Docs
- Pandas Docs
- Scikit-learn Docs
-

The references used in this project include official documentation and reliable learning resources related to Python and data science. The Python documentation was referred to for understanding core programming concepts and syntax. The Pandas documentation was extensively used for data cleaning, manipulation, and handling structured datasets efficiently. Additionally, the Scikit-learn documentation provided guidance on implementing machine learning algorithms, particularly Linear Regression, as well as evaluating model performance using metrics such as Mean Squared Error (MSE) and R^2 score. These sources ensured accurate implementation of techniques and a clear understanding of data analysis methodologies.

Linkdin link: https://www.linkedin.com/posts/gaurav-raj-48a027228_datascience-machinelearning-eda-ugcPost-7451912896848084992-42Hx?utm_source=share&utm_medium=member_desktop&rcm=ACoAADj00-oBUVm8NB-PZ5aGbJnybhSOH0DGB1Y

Google Drive link:

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